

Khovanov's Homology for Knots and Links

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1 Introduction

Given two links $\mathcal{L}_1$ and $\mathcal{L}_2$, it is very difficult to tell whether they are different links. Playing around with Reidemeister moves quickly becomes practically impossible as we increase the complexity of the links, and so we need to take a more clever approach. We use *link invariants*, mathematical objects that we assign to our links. If two links have different invariants, then they certainly can't be the same. The most basic examples are numerical invariants, like the linking number, where our mathematical objects are just numbers. More sophisticated examples are polynomial invariants, where our mathematical objects are polynomials. In this paper we are concerned with the polynomial invariant first “categorified” (see §7) by Mikhail Khovanov: the Jones polynomial.

Definition 1. Given a link diagram L , we define the *Kauffman bracket* $\langle L \rangle \in \mathbb{Z}[q^{\pm}]$ by the following axioms:

$$\langle \emptyset \rangle = 1, \quad \langle \bigcirc L \rangle = (q + q^{-1})\langle L \rangle, \quad \langle \nearrow \rangle = \langle \searrow \rangle - q\langle \downarrow \rangle.$$

The use of the letter “ q ” as a variable is an allusion to the fact that the Jones polynomial, defined below, is a *quantum* invariant of knots. Given an oriented link diagram L , a crossing of the form $\nearrow$ is called a *positive* crossing, and $\searrow$ is called a *negative* crossing. We set:

$$\begin{aligned} n_+ &= \text{number of positive crossings of } L, \\ n_- &= \text{number of negative crossings of } L. \end{aligned}$$

The Kauffman bracket is not a link invariant. However, the *unnormalized Jones polynomial* $\hat{J}(L) = (-1)^{n_-} q^{n_+ - 2n_-} \langle L \rangle$ is a link invariant.

Polynomials have a lot of nice structure, but we would like to use a type of link invariant with even more structure. Fortunately, Khovanov has introduced a very nice example of such a link invariant. Not only is it more powerful than the Jones polynomial (in fact, as we will see in §5, it is an extension of the Jones polynomial), but it also can be carried along “maps” between links (these are the 4-dimensional cobordisms of §6), which is very useful in applications.

The purpose of this paper is to give an introduction to Khovanov homology. We mostly follow the constructions given in [B-N05]. For Khovanov's original paper, see [K99].

2 Some Definitions

The following notions will be essential in the subsequent discussion.

2.1 Manifolds and Cobordisms

Intuitively, a *manifold* is a topological space with all the properties that make a space “nice.” Manifolds have a well-defined dimension, they look like Euclidean space when you zoom in on them, and for dimensions 1 and 2, they are just curves and surfaces.

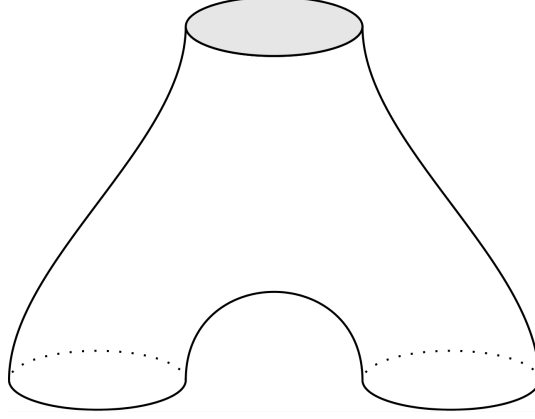


Figure 1: A pair of pants

Definition 2. An n -dimensional manifold-with-boundary M is a second countable Hausdorff space such that for every $p \in M$, either

- (i) p has an open neighborhood $U \subseteq M$ homeomorphic to $\mathbb{R}^n$, or
- (ii) p has an open neighborhood $U \subseteq M$ homeomorphic to $\{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_1 \geq 0\}$, where the homeomorphism sends p to a point with $x_1 = 0$.

The collection of $p \in M$ satisfying (ii), the *boundary* of M , is denoted ∂M . If $\partial M = \emptyset$, then M is called a *manifold*, or *n-manifold* if we want to specify its dimension. For convenience, we say that $\emptyset$ is an n -manifold for any n .

We will be concerning ourselves with *cobordisms*: manifolds that connect two other manifolds together.

Definition 3. Given two compact n -dimensional manifolds M, N , a *cobordism* between M and N is an $(n+1)$ -dimensional manifold-with-boundary X whose boundary is the disjoint union $M \cup N$.

As an example, consider the cobordism in Figure 1 between a single circle and a disjoint union of two circles. We call this a “pair of pants.” Note that there is a saddle point where the two legs branch off.

2.2 Chain Complexes and Homology

The key insight of algebraic topology is that we can use algebraic structures to encode topological information. We will be concerned with the following algebraic structures.

Definition 4. A *graded vector space* is a vector space W with a specified decomposition $W = \bigoplus_{m \in \mathbb{Z}} W_m$. The *graded dimension* of W is defined as $\text{grdim } W := \sum_{m \in \mathbb{Z}} q^m \dim W_m$.

Definition 5. The *tensor product* $V \otimes W$ of vector spaces V and W with bases $\{v_i\}_{i=1}^n$ and $\{w_j\}_{j=1}^m$ is the vector space with basis $\{v_i \otimes w_j\}_{i=1, j=1}^{n, m}$. For the purposes of this paper, it suffices to think of the $v_i \otimes w_j$ as merely formal symbols.

Definition 6. A *chain complex* $\mathcal{C}$ consists of a collection $\{\mathcal{C}^i\}_{i \in \mathbb{Z}}$ of (possibly graded) vector spaces and linear maps $d_i : \mathcal{C}^i \rightarrow \mathcal{C}^{i+1}$ (called *differentials*) with the property $d_{i+1} \circ d_i = 0$. We denote such a complex with the following diagram, where arrows denote maps:

$$\dots \xrightarrow{d_{i-2}} \mathcal{C}^{i-1} \xrightarrow{d_{i-1}} \mathcal{C}^i \xrightarrow{d_i} \mathcal{C}^{i+1} \xrightarrow{d_{i+1}} \dots$$

A *chain map* between two complexes $\mathcal{C}$ and $\mathcal{D}$ is collection $\{F_i\}_{i \in \mathbb{Z}}$ of linear maps $F_i : \mathcal{C}^i \rightarrow \mathcal{D}^i$ such that $F_i \circ d_i = d_i \circ F_{i+1}$, where the first d_i acts on $\mathcal{D}^i$ and the second acts on $\mathcal{C}^i$. When we refer to some generic d_i , F_i , we often simply write d , F . Therefore, the previous equation is just $F \circ d = d \circ F$.

Since $d_{i+1} \circ d_i = 0$, we have $\text{im } d_i \subseteq \ker d_{i+1}$, and so we can make the following definition.

Definition 7. The *homology* $\mathcal{H}(\mathcal{C})$ of a chain complex $\mathcal{C}$ is the collection $\{\mathcal{H}^i(\mathcal{C})\}_{i \in \mathbb{Z}}$, where $\mathcal{H}^i(\mathcal{C}) := \ker d_{i+1} / \text{im } d_i$. If the $\mathcal{C}^i$ are graded, then each $\mathcal{H}^i(\mathcal{C})$ is again a graded vector space, and so we define the *graded Euler characteristic* of $\mathcal{C}$ to be

$$\chi(\mathcal{C}) := \sum_{i \in \mathbb{Z}} (-1)^i \text{grdim } \mathcal{H}^i(\mathcal{C}).$$

Finally, we introduce a way in which chain maps may be seen as equivalent.

Definition 8. Given chain complexes $\mathcal{C}$ and $\mathcal{D}$ and two chain maps F and G between them, we say that F and G are *chain homotopic* if there exist maps $\{h_i\}_{i \in \mathbb{Z}}$, $h_i : \mathcal{C}^i \rightarrow \mathcal{D}^{i-1}$ such that $F_i - G_i = h_{i+1} \circ d_i + d_{i-1} \circ h_i$. We call h a *chain homotopy*.

2.3 Categories and Functors

The language of category theory allows us to talk precisely about correspondences between different kinds of mathematical objects.

Definition 9. A *category* $\mathbf{C}$ consists of a formal collection (not a set) of *objects* and a collection of *morphisms* between these objects. For a morphism f from an object A to an object B , we write $f : A \rightarrow B$. We suppose that we can compose morphisms associatively, and that every object A has an identity morphism 1_A that satisfies $f \circ 1_A = f$ for all $f : A \rightarrow B$ and $1_A \circ g = g$ for all $g : B \rightarrow A$, where B is an arbitrary object.

A simple example of a category is **Grp**, whose objects are groups and whose morphisms are group homomorphisms. Note that morphisms need not be functions. For example, we will be interested in the category **2Cob**, whose objects are compact 1-manifolds (i.e. disjoint unions of circles) and whose morphisms are cobordisms between compact 1-manifolds. To compose two cobordisms, we just stack them on top of each other.

Now we define a *functor*, which we can think of intuitively as a “function” between two categories.

Definition 10. A *functor* $\mathcal{F}$ from a category $\mathbf{C}$ to a category $\mathbf{D}$ is a formal symbol such that for every object A of $\mathbf{C}$, $\mathcal{F}A$ is an object of $\mathbf{D}$, and for every morphism f in $\mathbf{C}$, $\mathcal{F}(f)$ is a morphism in $\mathbf{D}$. Furthermore, we require that $\mathcal{F}(f \circ g) = \mathcal{F}(f) \circ \mathcal{F}(g)$ for all morphisms f, g in $\mathbf{C}$, and that $\mathcal{F}(1_A) = 1_{\mathcal{F}A}$ for every object A of $\mathbf{C}$.

As a simple example, let **AGrp** be the category whose objects are abelian groups and whose morphisms are group homomorphisms. Then there is a functor $\mathcal{A}$ from **Grp** to **AGrp** such that for any group G in **Grp**, the object $\mathcal{A}G$ is the abelianization of G , and for any homomorphism $\varphi : G \rightarrow H$ in **Grp**, the morphism $\mathcal{A}\varphi : \mathcal{A}G \rightarrow \mathcal{A}H$ is just the map induced by φ .

Finally, we consider a notion that is very useful for making arguments about morphisms.

Definition 11. A *commutative diagram* in a category $\mathbf{C}$ is a diagram that depicts objects of $\mathbf{C}$ and arrows between them, where the arrows are labeled with morphisms in $\mathbf{C}$. Furthermore, we require that if one can get from object A to object B in the diagram by traversing two different sequences $A \xrightarrow{f_1} \dots \xrightarrow{f_n} B$, $A \xrightarrow{g_1} \dots \xrightarrow{g_m} B$ of arrows, then we must have $f_n \circ f_{n-1} \circ \dots \circ f_1 = g_m \circ g_{m-1} \circ \dots \circ g_1$.

As an example, if the following is a commutative diagram, then we have $f = h \circ g$.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow g & \nearrow h & \\ C & & \end{array}$$

3 A Formal Construction on Link Diagrams

Given an oriented link diagram L , we can obtain a nonsingular diagram by “smoothing” each crossing of the diagram. We do this by replacing each crossing $\times$ with either $\smile$, a *0-smoothing*, or $\succ$, a *1-smoothing*. If a diagram has n crossings, then there will be 2^n different ways to smooth the diagram, and $\binom{n}{i}$ to smooth the diagram with exactly i 1-smoothings. Therefore, as in Figure 2, we may form a “cube” of smoothings of L aligned so that it appears as a sequence of vertical columns where the i th column contains all the smoothings that use exactly i 1-smoothings. We label the edge of the cube between any two smoothings that differ by precisely one 1-smoothing with a cobordism between these smoothings, where the cobordism is made up of cylinders between circles not involved in the particular 1-smoothing, and is either a pair of pants or an upside-down pair of pants between the circles involved in the 1-smoothing (see Figure 2).

Recall from §2.3 that **2Cob** is the category of compact 1-manifolds whose morphisms are cobordisms. Each smoothing is an object of **2Cob**, and each cobordism between smoothings is a morphism in **2Cob**. Therefore, each vertex in the cube of smoothings is an object of **2Cob**, and each edge in the cube of smoothings is a morphism in **2Cob**. In fact, we have the following lemma.

Lemma 12. *The cube of smoothings of a link diagram L is a commutative diagram.*

Proof. The cube is certainly a diagram depicting objects and arrows (the edges of the cube) between them, and these arrows are labeled with morphisms. Therefore, it remains to show that any two traversals of arrows produces the same morphism. Given smoothings A and B , where A has fewer 1-crossings than B , a traversal of arrows from A to B corresponds to a stacking of cobordisms. These cobordisms are successive pairs of pants (and upside-down pairs of pants) with saddle points in every place where A and B differ by a 0- or 1-smoothing. Depending on the sequence of cobordisms, some saddle points are higher up than others, but we can deform the cobordism by raising and lowering the saddle points as necessary to get them all to the same height. This resulting cobordism may be obtained no matter which sequence of cobordisms we started with. That is to say, given any sequence from A to B of morphisms in the cube of smoothings, the composition of all the morphisms in the sequence is independent of the sequence we choose. This precisely means that the cube of smoothings is a commutative diagram. $\square$

Over each vertical column of our cube of smoothings, we take the formal direct sum of the smoothings in the column. This is no longer an object of **2Cob**, since there is *a priori* no way to take direct sums in **2Cob**. Therefore, we let **Mat(2Cob)** denote the category whose objects are such direct sums, and whose morphisms are formal matrices whose entries are morphisms of **2Cob**. The benefit of this procedure is that we have now “collapsed” Figure 2 down to a complex (where $S_1^i, \dots, S_{\binom{n}{i}}^i$ is some ordering of the smoothings of L with i 1-smoothings):

$$\dots \xrightarrow{d_{i-2}} \bigoplus_{j=1}^{\binom{n}{i-1}} S_j^{i-1} \xrightarrow{d_{i-1}} \bigoplus_{j=1}^{\binom{n}{i}} S_j^i \xrightarrow{d_i} \bigoplus_{j=1}^{\binom{n}{i+1}} S_j^{i+1} \xrightarrow{d_{i+1}} \dots$$

Let us denote this complex by $\llbracket L \rrbracket$. The maps d_i are matrices whose $j\ell$ th entry is the cobordism on the edge from S_ℓ^i to S_j^{i+1} (if there is no such edge, then this entry is just 0). But this is not a differential, since we certainly do not have $d_i \circ d_{i-1} = 0$. This can be fixed fairly easily, though.

We know by Lemma 12 that the *faces* of the cube commute (i.e. they are commutative diagrams), and so if we add a minus sign to precisely one edge of every face (see Figure 2), then traversing the face one way will give some morphism f , and traversing the other way will give $-f$. Now, every entry of the matrix $d_i \circ d_{i-1}$ is the sum of both ways of traversing some face of the cube. If we now include our minus signs, then this sum is now of the form $f - f = 0$, and so $d_i \circ d_{i-1}$ is the zero matrix, as desired. Now the d_i are truly differentials.

What we have constructed looks exactly like a chain complex, but its entries are not algebraic objects at all. We will call this a *formal chain complex*, and we will consider the category **Com(Mat(2Cob))** whose objects are these formal chain complexes, and whose morphisms are chain maps $\{F_i\}_{i \in \mathbb{Z}}$ between them,

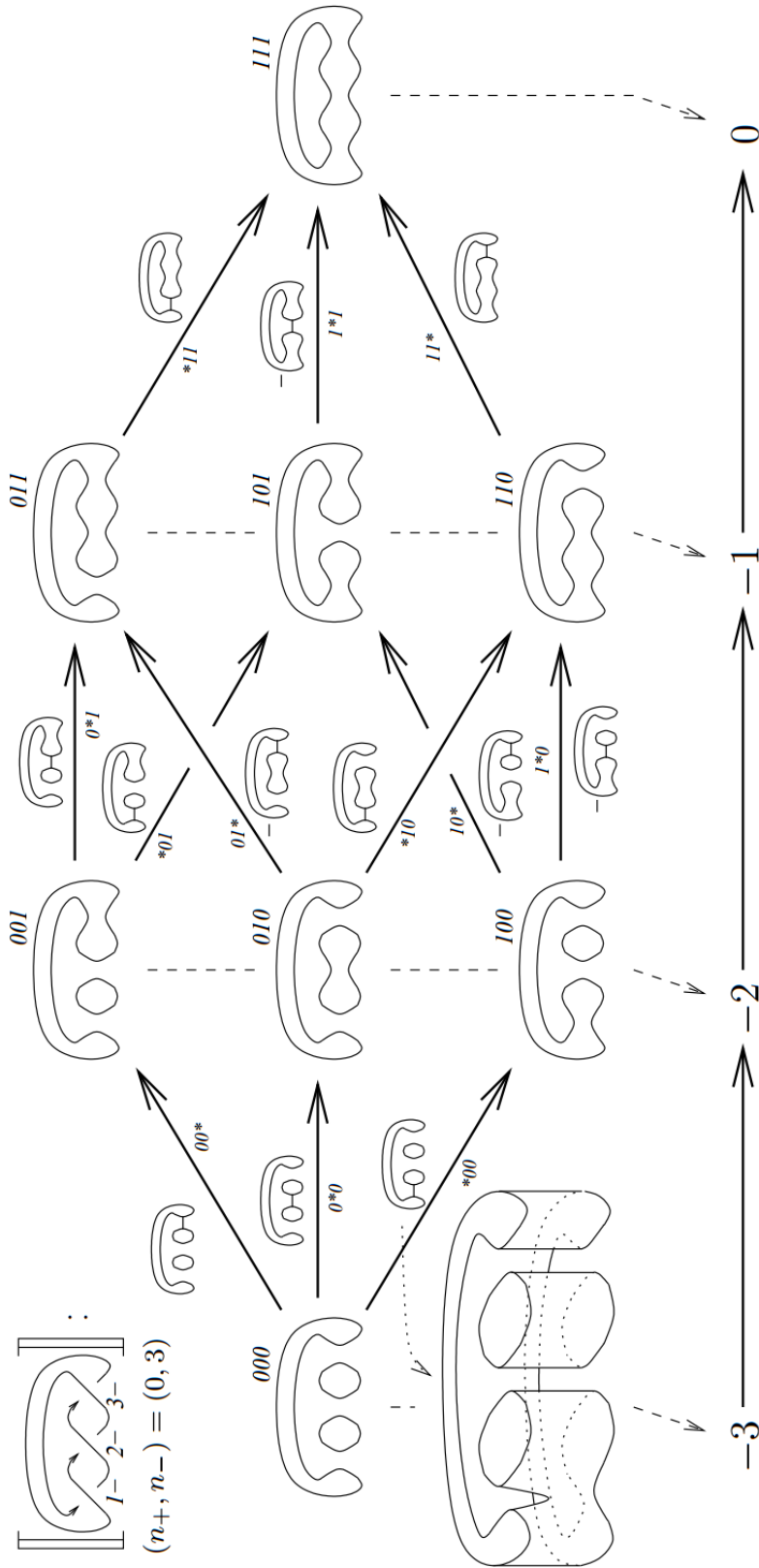


Figure 2: (Created by Dror Bar-Natan; see Figure 1 in [B-N05]) The cube of smoothings of a trefoil

where each F_i is a morphism in $\mathbf{Mat}(\mathbf{2Cob})$. It is now our goal to obtain from this an actual chain complex of vector spaces.

4 TQFTs

The goal of this section is to describe a coherent way to associate vector spaces with smoothings of a link diagram, and linear maps with cobordisms of smoothings. The acronym “TQFT” stands for “Topological Quantum Field Theory.” This name is due to the physical motivation with which TQFTs were defined.

Definition 13. (Cf. §2 of [A88]) Fix a field k , and let $\mathbf{Vect}_k$ denote the category of vector spaces over k whose morphisms are linear maps. A *TQFT* is a functor $\mathcal{F}$ from $\mathbf{2Cob}$ to $\mathbf{Vect}_k$ such that, if $M \cup N$ is a disjoint union of compact 1-manifolds, then $\mathcal{F}(M \cup N) = \mathcal{F}M \otimes \mathcal{F}N$.

Observe that all cobordisms in $\mathbf{2Cob}$ can be built up using the following four kinds of “building blocks”: A cup $\ominus$, a cap $\oplus$, a pair of pants $\Join$, and an upside-down pair of pants Υ . We have excluded the cylinder, since this is a more-or-less trivial building block. Now, in order to specify a TQFT $\mathcal{F}$, it suffices to say what vector space V is assigned to the circle $\bigcirc$, since all compact 1-manifolds are disjoint unions of circles, and to say what linear maps are assigned to the above building blocks. Since $\mathcal{F}$ is a functor and the cylinder is the identity morphism from a circle to a circle, the $\mathcal{F}$ must take the cylinder to the identity map on V . Also, we read cobordisms from top to bottom, so that $\mathcal{F}(\Upsilon)$ is a map from $V \oplus V$ to V .

Definition 14. Let $k = \mathbb{C}$, and let V be a 2-dimensional k -vector space with basis $\{v_+, v_-\}$. We will denote by $\mathcal{F}$ the TQFT satisfying $\mathcal{F}(\bigcirc) = V$ and:

$$\begin{aligned} \mathcal{F}(\Upsilon) &= \mu : V \otimes V \rightarrow V, & \mu : \begin{cases} v_+ \otimes v_- \mapsto v_- & v_+ \otimes v_+ \mapsto v_+ \\ v_- \otimes v_+ \mapsto v_- & v_- \otimes v_- \mapsto 0 \end{cases} \\ \mathcal{F}(\Join) &= \Delta : V \rightarrow V \otimes V, & \Delta : \begin{cases} v_+ \mapsto v_+ \otimes v_- + v_- \otimes v_+ \\ v_- \mapsto v_- \otimes v_- \end{cases} \\ \mathcal{F}(\ominus) &= \varepsilon : V \rightarrow F, & \varepsilon : \begin{cases} v_+ \mapsto 0 & v_- \mapsto 1 \end{cases} \\ \mathcal{F}(\oplus) &= \eta : F \rightarrow V, & \eta : \begin{cases} 1 \mapsto v_+ \end{cases} \end{aligned}$$

We will want to allow formal sums of morphisms to also count as morphisms in $\mathbf{2Cob}$. We will want to make a further slight change, due to the following lemma. Note that a sphere is a cap $\oplus$ followed by a cup $\ominus$, and that a torus is a similar composition of building blocks, as indicated by the formula, so that these names make sense. The 4 Tubes condition may be viewed as merely another technical condition.

Lemma 15. *The maps μ , Δ , ε , and η satisfy Bar-Natan’s Conditions, which are as follows:*

$$\begin{aligned} \varepsilon \circ \eta(1) &= 0 & (\text{Sphere}) \\ \varepsilon \circ \mu \circ \Delta \circ \eta(1) &= 2 & (\text{Torus}) \\ (\eta \otimes \eta) \circ \varepsilon \circ \mu + \Delta \circ \eta \circ (\varepsilon \otimes \varepsilon) &= \text{Id} \otimes (\eta \circ \varepsilon) + (\eta \circ \varepsilon) \otimes \text{Id} & (4 \text{ Tubes}) \end{aligned}$$

Proof. By definition, $\varepsilon \circ \eta(1) = \varepsilon(v_+) = 0$, and we also compute:

$$\begin{aligned} \varepsilon \circ \mu \circ \Delta \circ \eta(1) &= \varepsilon \circ \mu \circ \Delta(v_+) \\ &= \varepsilon \circ \mu(v_+ \otimes v_- + v_- \otimes v_+) \\ &= \varepsilon(2v_-) \\ &= 2. \end{aligned}$$

The verification of the 4 Tubes condition is just as straightforward and is therefore omitted. $\square$

The Sphere condition tells us that if a cobordism X has a sphere as a connected component, then $\mathcal{F}$ does not distinguish between X and the 0 morphism (i.e. the empty formal sum). The Torus condition tells us that if a cobordism X has a torus as a connected component, then $\mathcal{F}$ does not distinguish between X and the cobordism obtained by removing the torus from X , provided that we multiply the latter by 2. The 4 Tubes condition also has a straightforward, albeit more complicated, topological interpretation. Since $\mathcal{F}$ makes no distinction between cobordisms identified via Bar-Natan's conditions, we may as well suppose that such cobordisms are identical morphisms. That is, instead of considering the category $\mathbf{2Cob}$, we instead consider the category $\mathbf{2Cob}_{/\ell}$, the quotient of $\mathbf{2Cob}$ by these relations.

It is good that we are using $\mathbf{2Cob}_{/\ell}$ instead of $\mathbf{2Cob}_{/\ell}$, because it takes us one step closer to obtaining an actual link invariant. More precisely, consider the category $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$, whose objects are the objects of $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))$, and whose morphisms are equivalence classes of morphisms of $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))$ modulo chain homotopy. We now have the following theorem.

Theorem 16. *Given a link $\mathcal{L}$ with link diagram L , the isomorphism class of $\llbracket L \rrbracket$ in $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ is an invariant of $\mathcal{L}$. In particular, $\llbracket L \rrbracket$ is invariant under Reidemeister moves applied to the link diagram L .*

The proof is rather technical and is quite beyond the scope of this expository paper. For a proof, see Theorem 1 in [B-N05].

We can extend $\mathcal{F}$ by linearity to $\mathbf{Mat}(\mathbf{2Cob}_{/\ell})$. For example, we have $\mathcal{F}((\bigcirc \cup \bigcirc) \oplus \bigcirc) = (V \otimes V) \oplus V$. We can also extend $\mathcal{F}$ to $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ to obtain actual chain complexes of vector spaces. We have thus found a way to associate a chain complex of vector spaces to any link $\mathcal{L}$: If L is a link diagram of $\mathcal{L}$, then the chain complex we are looking for is $\mathcal{F}\llbracket L \rrbracket$, an object of $\mathbf{Com}(\mathbf{Vect}_k)_{/h}$. It follows from Theorem 16 that $\mathcal{F}\llbracket L \rrbracket$ is an invariant of $\mathcal{L}$.

5 Gradings and the Jones Polynomial

We consider the vector space V defined in Definition 14 as a graded vector space $V = kv_+ \oplus kv_-$, where we let kv_+ have a grading of 1, and let kv_- have a grading of -1 (that is to say, $V = V_1 \oplus V_{-1} = kv_+ \oplus kv_-$). Note that $\text{grdim } V = q + q^{-1} = \langle \bigcirc \rangle$, where $\langle \cdot \rangle$ is the Kauffman bracket defined in Definition 1. We now define a technical operation on graded vector spaces.

Definition 17. Given a graded vector space $W = \bigoplus_{m \in \mathbb{Z}} W_m$, we define the *degree shift* $W\{\ell\} := \bigoplus_{m \in \mathbb{Z}} W\{\ell\}_m$, where $W\{\ell\}_m = W_{m-\ell}$.

For example, $V\{1\} = V\{1\}_0 \oplus V\{1\}_2$, with $V\{1\}_0 = V_{-1} = kv_-$ and $V\{1\}_2 = V_1 = kv_+$. Given our functor $\mathcal{F}$ from $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ to $\mathbf{Com}(\mathbf{Vect}_k)_{/h}$, we define a functor $\mathcal{F}'$ from $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ to $\mathbf{Com}(\mathbf{Vect}_k)_{/h}$ given by

$$(\mathcal{F}'\llbracket L \rrbracket)^i := (\mathcal{F}\llbracket L \rrbracket)^{i+n_-} \{n_+ - 2n_-\},$$

where n_+ and n_- were defined in Definition 1. Given a link $\mathcal{L}$ with link diagram L , we call the homology of $\mathcal{F}'\llbracket L \rrbracket$ the *Khovanov homology* of $\mathcal{L}$. Notice that the modifications we made to $\mathcal{F}\llbracket L \rrbracket$ using n_+ and n_- to obtain $\mathcal{F}'\llbracket L \rrbracket$ are the same as the modifications we made to $\langle L \rangle$ to obtain $\widehat{J}(L)$ in Definition 1. This is done so that we may have the following theorem (cf. Theorem 1 in [B-N02]).

Theorem 18. *Given a link diagram L , we have $\chi(\mathcal{F}'\llbracket L \rrbracket) = \widehat{J}(L)$.*

Explicitly computing both $\chi(\mathcal{F}'\llbracket L \rrbracket)$ and $\widehat{J}(L)$ in the case where L depicts the Hopf link should suffice to convince the reader that Theorem 18 is true. Theorem 18 is the precise statement of the claim we made in §1 that Khovanov homology extends the Jones polynomial.

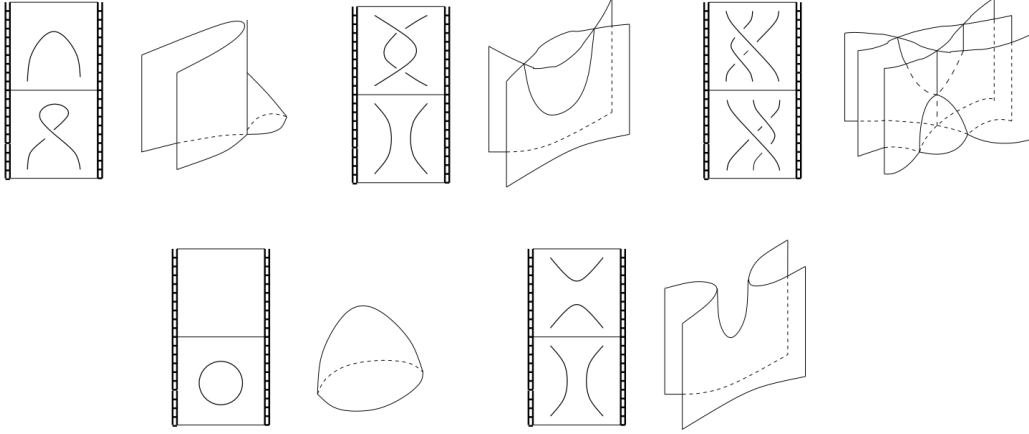


Figure 3: (Created by Dror Bar-Natan; see Figure 10 in [B-N05]) Building blocks for movies

6 4-Dimensional Cobordisms and Chain Maps

We have a way of obtaining formal chain complexes from links. We would also like to have a way of obtaining chain maps between these chain complexes. This section elides more details than the previous sections, but the goal is to nevertheless give an overview of the ideas of the theory.

Links live in $\mathbb{R}^3$, and so a cobordism between links will be a 2-surface living in $\mathbb{R}^4$. We can think of link diagrams as pairings (L, π) , where $L \subseteq \mathbb{R}^3$ is a link, and π is our chosen projection onto a plane. Therefore, we can think of cobordisms between link diagrams as cobordisms between links living in $\mathbb{R}^3 \times [0, 1]$, where the last parameter is viewed as “time,” and at each time t we have a projection π_t . Looking through the lens of π_t as time goes by, we see a “movie” of link diagrams. Note that these cobordisms are different from the morphisms of $\mathbf{2Cob}$, which live in $\mathbb{R}^3$ rather than $\mathbb{R}^4$. The singular 2-surfaces shown in Figure 3 are only projections onto $\mathbb{R}^3$ of nonsingular 2-cobordisms that live in $\mathbb{R}^4$.

Formally, we let $\mathbf{Cob}^4$ denote the category whose objects are link diagrams, and whose morphisms are movies (i.e. cobordisms in $\mathbb{R}^4$ between link diagrams), and let $\mathbf{Cob}_{/i}^4$ denote the quotient of $\mathbf{Cob}^4$ by isotopies (we take the quotient in the same way as we took the quotient $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ of $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))$ in §4). We would like to have a functor $\mathcal{G}$ from $\mathbf{Cob}_{/i}^4$ to $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$.

Each movie in $\mathbf{Cob}^4$ can be built by elementary building blocks in the same way that cobordisms in $\mathbf{2Cob}$ can be built by cups, caps, pairs of pants, and upside-down pairs of pants. In fact, there are five building blocks: three building blocks are the movies that depict the performance of a Reidemeister move, one building block depicts a cup/cap (we can watch movies either forwards or on rewind), or a pair of pants/upside-down pair of pants (See Figure 3).

Just as we defined μ , Δ , ε , and η from the building blocks of $\mathbf{2Cob}$, we can define morphisms in $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))$ from these building blocks. The precise construction of these morphisms is beyond the scope of this paper, so suffice it to say that this procedure is morally similar to the procedure in the case of $\mathbf{2Cob}$. See §8.1 of [B-N05] for the details of the construction. Once we have constructed these morphisms, we have a functor $\mathcal{G}'$ from $\mathbf{Cob}^4$ to $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))$. To check that this actually gives us a functor $\mathcal{G}$ from $\mathbf{Cob}_{/i}^4$ to $\mathbf{Com}(\mathbf{Mat}(\mathbf{2Cob}_{/\ell}))_{/h}$ requires that we check that our construction is invariant under the “movie movies” of Carter and Saito, which are the analogues of Reidemeister moves in the case of 4-dimensional cobordisms. There are many of these, and the proof that our construction is invariant under movie moves can be found in §8.3 of [B-N05].

The existence of such a functor $\mathcal{G}$ is quite useful, and one application of this can be found in [R04], where the author uses a variant of $\mathcal{G}$ to obtain results on the slice genus of a knot.

7 Conclusion

With the Jones polynomial as our template, we have constructed a more nuanced algebraic link invariant that admits morphisms induced by cobordisms between links. In particular, we have constructed a homology theory whose Euler characteristic is the Jones polynomial. It is for this reason that Khovanov homology is referred to as a “categorification” of the Jones polynomial. Indeed, Khovanov homology is “stronger” than the Jones polynomial in the sense that there exist links with the same Jones polynomial, but different Khovanov homology. In fact, while it is unknown whether the Jones polynomial detects the unknot, it is known that Khovanov homology detects the unknot (in the sense that the unknot has a unique Khovanov homology).

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